

Expression of Interest

Artificial Intelligence for Climate Learning and Sustainable Development

Learning Water-Energy-Food Trade-offs with Transparent Agentic AI: A Scenario-Based Demonstration from Sleman, Indonesia

Zulfikar Aji Kusworo

Ministry of Energy and Mineral Resources of the Republic of Indonesia

Corresponding author: Zulfikar Aji Kusworo, zakusworo@esdm.go.id

Project repository: <https://github.com/zakusworo/wef-agentic>

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Abstract

Climate learning requires an understanding of how water, energy and food decisions interact, particularly where local data are incomplete and policy choices involve competing objectives. This chapter presents WEF-Agentic, a working scenario-based demonstrator situated in Sleman, Indonesia, for examining these interactions through transparent numerical modelling and AI-assisted interpretation.

The system computes monthly water balance, irrigation-related water stress, crop production and electricity demand before generating any narrative. Water, energy and food agents interpret the resulting quantities; a critic examines consistency, and a coordinator summarizes trade-offs. The interface exposes assumptions, provenance and warnings alongside five scenarios. Recorded runs preserve model identities and outputs, allowing readers to compare what the numerical model calculates with what the language models say.

The chapter will connect this workflow to practical climate-learning activities: tracing cross-sector effects, distinguishing observations from assumptions, comparing alternative futures, and questioning AI-generated recommendations. A reproducible software demonstration and documented scenario experiments will illustrate the approach. Synthetic verification data will be distinguished from reanalysis inputs, and uncalibrated parameters will be identified. The chapter will not infer learning gains or policy effectiveness from software operation alone.

The contribution is a practical account of how constrained AI interpretation can support exploration of sustainability trade-offs while keeping uncertainty and human judgment visible. It will discuss adaptation to other Global South settings, requirements for local validation, and a proposed evaluation of learners' ability to explain and challenge model-based claims.